

Reconstructing frustrated signs from amplitudes with fixed-node and Krylov updates

A. Otaifi¹

¹*Arnold Sommerfeld Center for Theoretical Physics,
Ludwig-Maximilians-Universität München, 80333 München, Germany*

(Dated: October 3, 2026)

We separate the sign and positive-amplitude parts of a real many-body wave function, $\psi(x) = s(x)a(x)$ with $a(x) = |\psi(x)| > 0$, in the frustrated square-lattice spin-1/2 J_1 - J_2 Heisenberg model. For a fixed amplitude, one shifted Krylov step gives $s'(x) = s(x) \text{sgn}[T - r_s(x)]$, where $r_s(x) = (H\psi)(x)/\psi(x)$. At $J_2/J_1 = 0.5$ this reduces the exact 4×4 probability weight on incorrectly signed configurations from 1.27×10^{-2} to 3.41×10^{-4} . On 6×6 , using exactly the same neural-network amplitude, the reconstructed signs give the same energy as the original neural-network signs within sampling uncertainty. Conversely, lattice fixed-node projection improves the positive amplitude while keeping the signs fixed; an exact 4×4 fixed-node/Krylov loop reaches the exact ground-state signs. The reconstructed signs remain useful on 8×8 . The remaining scaling problem is to infer from fixed-node Monte Carlo walkers a positive amplitude function that can be evaluated on arbitrary configurations. Maximum likelihood targets $a_{\text{new}}^2 \propto a\phi_{\text{FN}}$; stochastic reconfiguration makes this update transfer between independent walker samples, but repeated 8×8 iterations do not yet lower the fixed-node energy reproducibly.

I. INTRODUCTION

The Monte Carlo sign problem is severe in frustrated spin systems [1, 2]. We use the square-lattice J_1 - J_2 Heisenberg model as a benchmark. At $J_2 = 0$ its ground-state signs obey the Marshall rule [3]; near $J_2/J_1 = 1/2$ that rule fails and accurate variational states become nontrivial [4, 5].

We ask whether signs and positive amplitudes can be improved separately. For a real wave function we write

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) = |\psi(x)| > 0. \quad (1)$$

Given a , a single Krylov step updates the signs. Given s , lattice fixed node gives a sign-free positive-amplitude problem [6]. Section II defines both updates. Section III tests them in order of increasing difficulty: first the sign update alone with an exact amplitude, then the exact feedback loop, then larger lattices, and finally the open problem of learning the amplitude from fixed-node walkers.

II. METHODS

A. Model and error measures

We study

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

on periodic lattices with $J_1 = 1$ in the $S^z = 0$ sector. For a spin configuration x , the Marshall rule is $s_M(x) = (-1)^{N_{\downarrow}^A(x)}$ up to an overall sign, where $N_{\downarrow}^A$ counts down spins on one checkerboard sublattice.

When the exact ground state $\psi_0 = s_0 a_0$ is available, we fix the irrelevant overall sign and measure the sign

error by the wrong-sign probability

$$w_s = \sum_{x: s(x) \neq s_0(x)} a_0(x)^2, \quad (3)$$

and the energy error by $\epsilon_{\text{rel}} = (E - E_0)/|E_0|$.

B. One-step Krylov sign update

For $\psi = sa$, define the local energy

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (4)$$

Because $[(T - H)\psi](x) = [T - r_s(x)]\psi(x)$, the sign of the shifted Krylov vector is

$$s_K(x) = s(x) \text{sgn}[T - r_s(x)] : \quad (5)$$

the sign of every configuration with $r_s(x) > T$ is flipped. Only this sign is kept; the amplitude factor $|T - r_s(x)|$ is discarded, so the method is not repeated Lanczos multiplication. The threshold T never uses the target signs. For exact systems, T minimizes the variational energy of $a(x)s_K(x)$ over all thresholds. For sampled systems, T is obtained from a two-cluster split of the sampled $r_s(x)$ values.

C. Fixed-node amplitude update

For a trial wave function ψ_T , an off-diagonal matrix element connecting x and y is sign violating when $H_{xy}\psi_T(y)/\psi_T(x) > 0$. Lattice fixed node removes such terms and adds their contribution to the diagonal [6]. The resulting Hamiltonian is sign free, keeps the signs of ψ_T fixed, and its ground-state energy E_{FN} is a variational upper bound. Let $\phi_{\text{FN}}(x) > 0$ be its positive

ground-state amplitude. In an exact small-system calculation, ϕ_{FN} replaces a directly before the next application of Eq. (5).

D. Learning the amplitude from walkers

In a walker calculation ϕ_{FN} is not available as a function. The stationary walker distribution is instead the mixed distribution

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (6)$$

If a positive model is trained by maximum likelihood with $p_\theta(x) \propto a_\theta(x)^2$, the exact maximum-likelihood target is

$$a_{\text{new}}^2(x) \propto a(x)\phi_{\text{FN}}(x), \quad (7)$$

the geometric mean of the current and fixed-node amplitudes. We precondition the likelihood gradient g with stochastic reconfiguration (SR),

$$\delta\theta = (S + \lambda I)^{-1}g, \quad (8)$$

where S is the quantum geometric tensor of $\log a_\theta$.

E. Numerical setup

The 4×4 calculations use the full $S^z = 0$ space of 12870 states; the second exact cluster is an $N = 20$ skew torus. For 6×6 and 8×8 we use a previously optimized translationally invariant vision-transformer neural quantum state (ViT) at $J_2/J_1 = 0.5$. Fixed-node energies are mixed estimates from walker populations of size M and projection length $\beta = 1.2$; 6×6 uses $M = 32$ and $M = 128$, 8×8 uses $M = 128$. Statistical conventions are summarized in Appendix B.

III. RESULTS

A. Benchmark 1: exact amplitude, Marshall signs, one Krylov step

The first test isolates the sign update. We take the exact 4×4 ground-state amplitude $a_0 = |\psi_0|$ from exact diagonalization and give it the Marshall signs, so the guide $\psi(x) = s_M(x)a_0(x)$ has a perfect amplitude and only its signs are wrong. For $J_2/J_1 \geq 0.8$, where the Marshall pattern is the wrong reference, we use the corresponding stripe sign rule instead. We apply Eq. (5) once and compare the result with exact diagonalization.

Figure 1(a) shows why one step works at $J_2/J_1 = 0.5$. The configurations on which the Marshall sign is wrong carry systematically larger local energies $r(x)$ than the correctly signed ones, and the energy-optimal threshold T separates the two groups: 98% of the wrong-sign probability lies at $r > T$. The wrong-sign probability therefore drops from 1.27×10^{-2} to 3.41×10^{-4} , a factor 37, and

the relative energy error from 2.19×10^{-2} to 1.04×10^{-3} , a factor 21. The improvement holds across the frustrated regime [Fig. 1(b,c)] and on the independent $N = 20$ cluster, where the factors are 50 and 24 [Fig. 1(d)].

B. Exact fixed-node-Krylov feedback

Next both parts are updated. We target $J_2/J_1 = 0.5$ but start from the exact $J_2 = 0$ amplitude with Marshall signs. Each iteration diagonalizes the current fixed-node Hamiltonian exactly, takes its positive ground state ϕ_{FN} as the new amplitude, and recomputes the signs with Eq. (5). No ground-state information enters either update; exact diagonalization of H is used only to evaluate the errors.

On $N = 16$ the wrong-sign probability falls from 1.27×10^{-2} to 8.85×10^{-7} after twenty iterations and to zero within numerical precision by iteration 100, where the amplitude fidelity is 0.9999946 and $\epsilon_{\text{rel}} = 3.08 \times 10^{-6}$ (Fig. 2). The $N = 20$ cluster behaves the same way: after 40 iterations $w_s = 2.55 \times 10^{-7}$, $1 - F_a = 8.15 \times 10^{-5}$, and $\epsilon_{\text{rel}} = 1.41 \times 10^{-5}$.

C. 6×6 : neural amplitude without an exact reference

On 6×6 we keep the ViT positive amplitude fixed and compare three sign choices: Marshall, one Krylov step from Marshall, and the original ViT signs. The ViT signs are a reference, not the exact ground state. One Krylov step reduces the $|\psi_{\text{ViT}}|^2$ -weighted disagreement with the ViT signs by a factor of about 20 on two independent samples [Fig. 3(a)]. With the same amplitude, the energies per site are $e_{\text{Marshall}} = -0.4982(16)$ and $e_{\text{ViT}} = -0.50345(16)$. Paired estimators on identical configurations give $e_{\text{Krylov}} - e_{\text{ViT}} = (1.2 \pm 2.9) \times 10^{-4}$, statistically zero, and $e_{\text{Krylov}} - e_{\text{Marshall}} = (-4.8 \pm 1.2) \times 10^{-3}$ [Fig. 3(b)].

The Krylov signs also improve fixed node. In a matched $M = 32$ comparison the Krylov-sign fixed-node energy is $E_{\text{FN}} = -18.128$ (-0.50356 per site), against $-17.973(30)$ (-0.49924 per site) for Marshall signs averaged over eight independent populations. Closing one loop iteration at $M = 128$ with a binned amplitude correction leaves the signs unchanged and gives a next amplitude correction statistically consistent with zero, with $E_{\text{FN}} = -18.1130(36)$ from two populations. Within this resolution the 6×6 loop has reached a fixed point.

D. 8×8 : fixed node with Krylov signs

On 8×8 the Krylov signs are applied to the ViT amplitude and used as the fixed-node guide. At $M = 128$

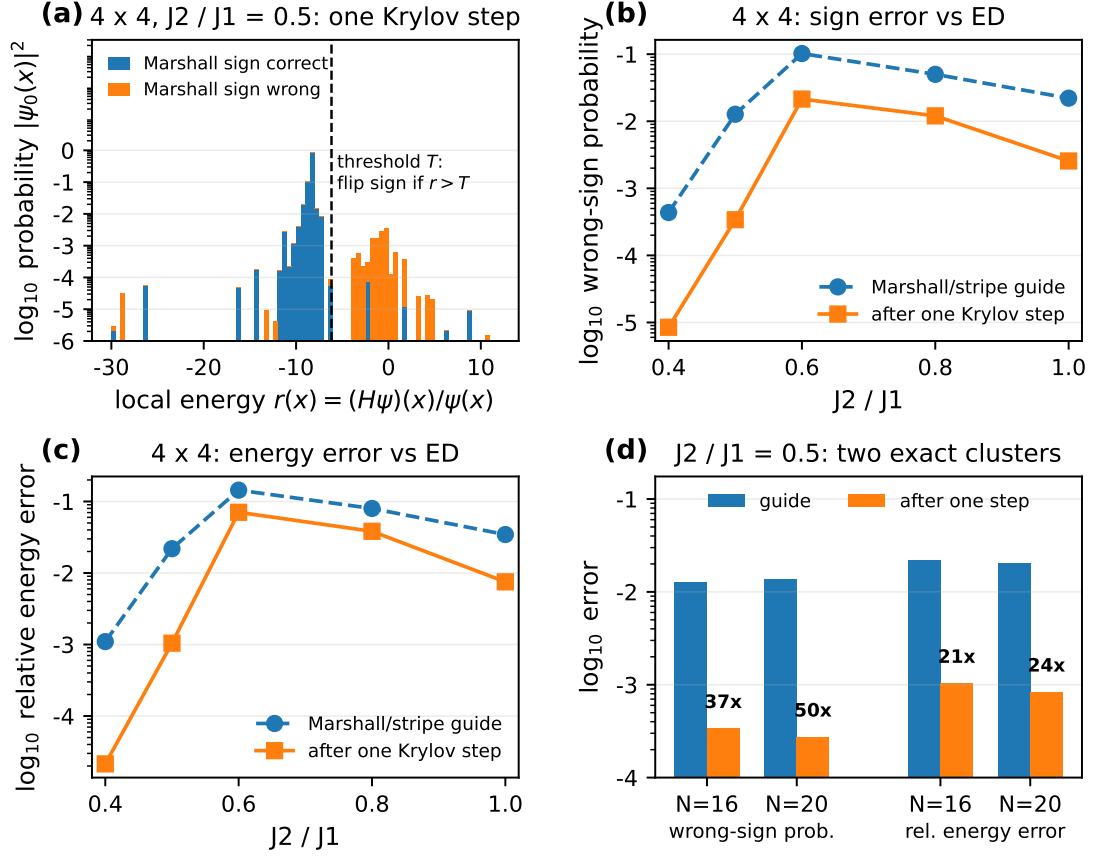


FIG. 1. Benchmark 1: one Krylov sign step on an exact amplitude. The guide is $\psi = s_M a_0$, with the exact positive amplitude a_0 and Marshall signs (stripe signs for $J_2/J_1 \geq 0.8$). (a) $|\psi_0|^2$ -weighted distribution of the local energy $r(x)$ at $J_2/J_1 = 0.5$, split into configurations where the Marshall sign is correct or wrong; the dashed line is the energy-optimal threshold T , and values outside the plotted range are collected in the end bins. (b) Wrong-sign probability and (c) relative energy error versus J_2/J_1 , before and after the step, compared with exact diagonalization. (d) The same comparison at $J_2/J_1 = 0.5$ on the $N = 16$ square and an $N = 20$ skew torus; labels give the reduction factors. All vertical axes are logarithmic and labelled by the base-10 exponent.

two independent populations give

$$\begin{aligned} E_{\text{FN}}^{\text{Krylov}} &= -31.8854 \quad (E/N = -0.498210), \\ E_{\text{FN}}^{\text{Marshall}} &= -31.6775 \quad (E/N = -0.494961). \end{aligned} \quad (9)$$

The two Krylov populations differ by 5×10^{-5} per site, the Marshall populations by 8×10^{-4} . Figure 4 compares these values with same-geometry results from Ref. [7]. The Krylov-sign value lies among the most accurate published estimates. Because only two populations at a single M are available, the population spread is a reproducibility scale, not an estimate of population-control bias.

E. Open problem: learning the fixed-node amplitude from walkers

To iterate the loop beyond exact systems, the walker information must be turned into a positive amplitude

that can be evaluated on any configuration. Equation (7) gives the target. On exact 4×4 , repeated half steps toward it steadily lower the energy, starting from either the exact amplitude or a uniform amplitude with Marshall signs [Fig. 5(a)].

On 8×8 the difficulty is statistical. The plain likelihood gradients from two independent walker populations have cosine -0.85 , so a gradient step fitted to one population worsens the other. With SR, Eq. (8) with $\lambda = 1$, the update fitted to one population raises the likelihood of the independent population for all step sizes up to $\eta = 0.01$, where the held-out gain is $+0.0156$ with importance-sampling effective sample size $3236/4096$ [Fig. 5(b)].

A transferable likelihood step is not yet an energy improvement. One SR iteration followed by a new Krylov step gave $E_{\text{FN}} = -31.9012$ on 8×8 , but an independent rebuild of the same update gave -31.839 , and a second iteration gave -31.815 , above the starting -31.885 . The 8×8 loop therefore does not yet converge. On exact

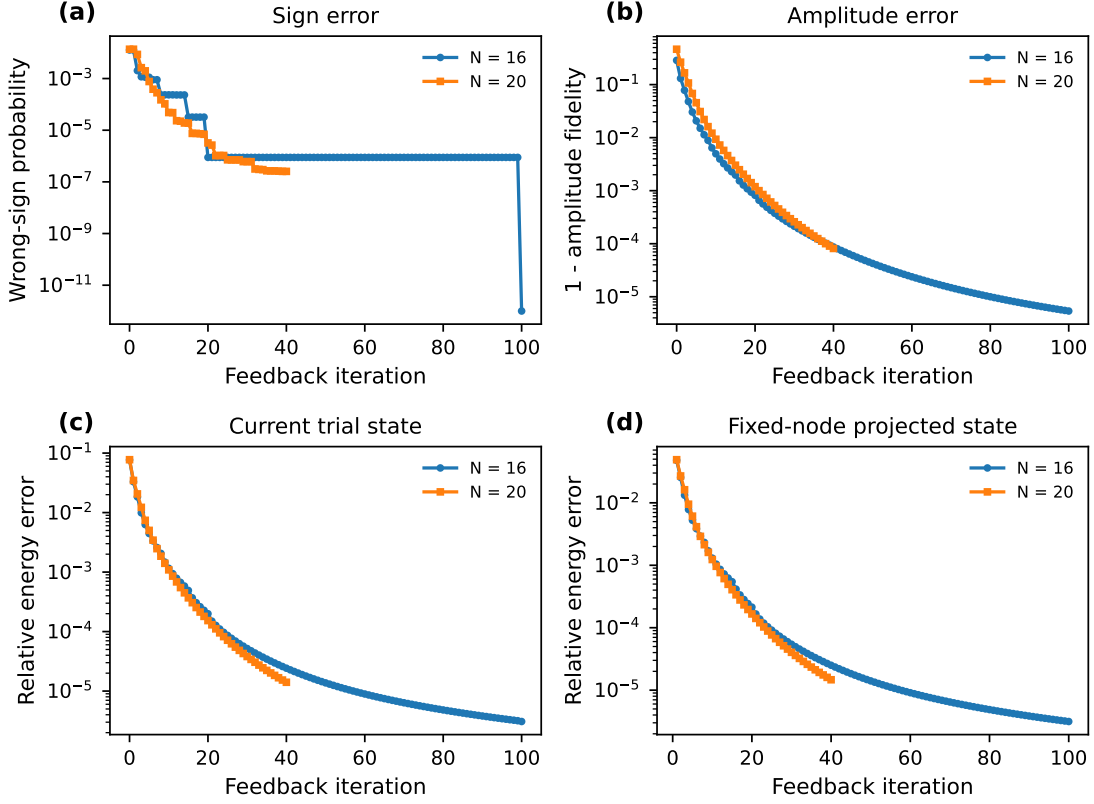


FIG. 2. Exact fixed-node/Krylov feedback at $J_2/J_1 = 0.5$ on $N = 16$ and $N = 20$, starting from the $J_2 = 0$ amplitude with Marshall signs. (a) Wrong-sign probability. (b) Amplitude error $1 - F_a$, with $F_a = |\langle a|a_0 \rangle|^2$. (c) Relative energy error of the updated trial state and (d) of the fixed-node projected state.

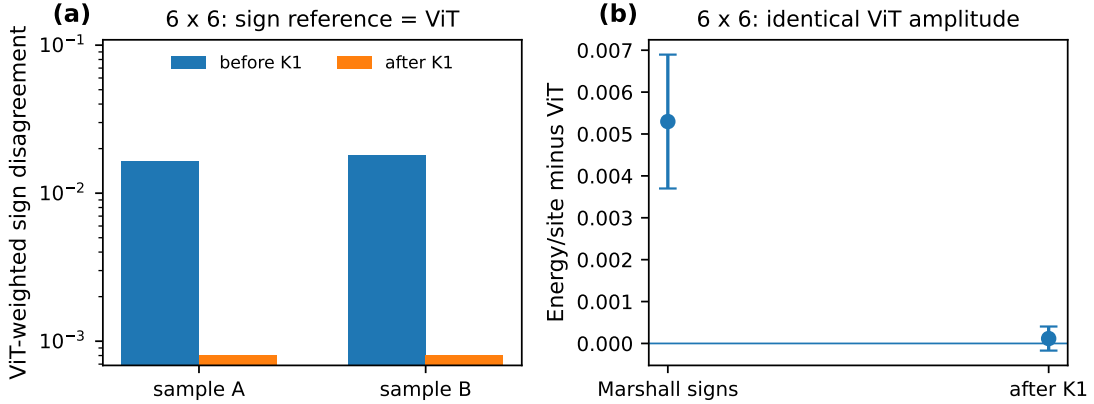


FIG. 3. 6×6 at $J_2/J_1 = 0.5$ with the ViT positive amplitude held fixed. (a) $|\psi_{\text{ViT}}|^2$ -weighted sign disagreement with the ViT signs before and after one Krylov step, on two independent samples. (b) Fixed-amplitude energy per site relative to the ViT-sign energy; error bars are between-chain standard errors.

4×4 , minimizing the frozen fixed-node energy with SR in a factorized network $a_\theta = a_k \exp[r_\theta]$ recovers 99.7% of the exact fixed-node energy gain, so the network can represent the update; the remaining question is the finite-sample estimator at scale.

IV. DISCUSSION

The results separate the two tasks. With a good positive amplitude, the dominant frustrated sign correction is captured by a single threshold in the local energy, reducing the exact sign error by factors of 37–50. Fixed node

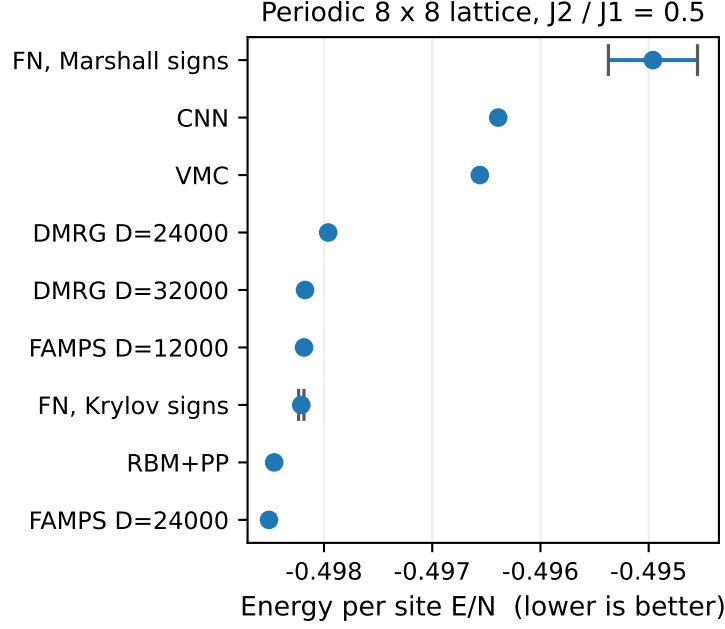


FIG. 4. Energy per site on the periodic 8×8 lattice at $J_2/J_1 = 0.5$. FN entries are lattice fixed node with Marshall or one-step Krylov signs; error bars are half the difference between two independent walker populations, and the grey ticks mark the individual populations. Literature values are from Ref. [7].

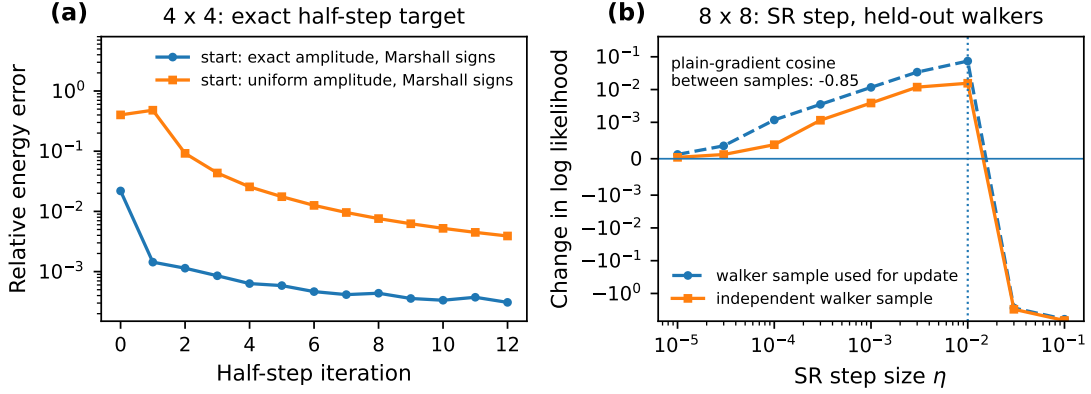


FIG. 5. Learning the positive amplitude. (a) Exact 4×4 iteration of the half-step target Eq. (7), starting from the exact or a uniform amplitude, both with Marshall signs. (b) 8×8 SR step-size scan: change in log likelihood on the walker population used for the update and on an independent population (symmetric-log axis).

solves the complementary positive-amplitude problem at fixed signs, and on exact clusters the two updates close into a loop that reaches the exact ground state. On 6×6 and 8×8 the Krylov signs reproduce the neural-network signs energetically and give competitive fixed-node energies.

The open problem is the amplitude step at scale: learning a callable positive amplitude from projector walkers. SR makes the likelihood update transfer between independent populations, but the resulting fixed-node energy is not yet reproducible over repeated iterations. The method also requires a useful starting amplitude and

should not be confused with repeated bare Krylov iteration, which is worse on 6×6 (Appendix A).

Appendix A: Direct second Krylov step

On 6×6 , retaining the full first Krylov amplitude gives $a_1(x) = |T_1 - r_0(x)|a_0(x)$ with coefficient of variation about 0.18. A second label-free threshold changes only 0.8–2.0% of the $|\psi_{\text{VIT}}|^2$ probability but raises the fixed-amplitude energy by 0.65–0.96. The feedback construction therefore re-solves the amplitude before another sign

update.

Appendix B: Statistical conventions

Exact-diagonalization data are deterministic up to eigensolver precision. For 6×6 fixed-amplitude energies we quote between-chain standard errors from 64 chains. For fixed-node data with two independent walker populations, error bars denote half the population difference and are reproducibility scales only; the 6×6 Marshall value uses the standard error over eight populations.

-
- [1] P. Henelius and A. W. Sandvik, Sign problem in Monte Carlo simulations of frustrated quantum spin systems, *Phys. Rev. B* **62**, 1102 (2000).
 - [2] M. Troyer and U.-J. Wiese, Computational complexity and fundamental limitations to fermionic quantum Monte Carlo simulations, *Phys. Rev. Lett.* **94**, 170201 (2005).
 - [3] W. Marshall, Antiferromagnetism, *Proc. R. Soc. Lond. A* **232**, 48 (1955).
 - [4] K. Choo, T. Neupert, and G. Carleo, Two-dimensional frustrated J_1 - J_2 model studied with neural network quantum states, *Phys. Rev. B* **100**, 125124 (2019).
 - [5] A. Chen and M. Heyl, Empowering deep neural quantum states through efficient optimization, *Nat. Phys.* **20**, 1476 (2024).
 - [6] H. J. M. van Bommel, D. F. B. ten Haaf, W. van Saarloos, J. M. J. van Leeuwen, and G. An, Fixed-node quantum Monte Carlo method for lattice fermions, *Phys. Rev. Lett.* **72**, 2442 (1994).
 - [7] X. Qian and M. Qin, Augmenting density matrix renormalization group with disentanglers, *Chin. Phys. Lett.* **40**, 057102 (2023).